The magnitude of the sodium atom's velocity change is

$$|u - V| = \frac{h}{M\lambda} \ ; \qquad (10)$$

Substituting the data gives

$$|u - V| = 2.9 \times 10^{-2}\ \mathrm{m \cdot s^{-1}} \ . \qquad (11)$$

Method 2. Sodium's excited $P_{3/2}$ to ground $S_{1/2}$ transition emits frequency

$$v_0 = \frac{c}{\lambda_0} \ ; \qquad (1)$$

The incident laser frequency is

$$v = \frac{c}{\lambda} \ . \qquad (2)$$

For sodium, relative to positive z, motion makes angle θ. Momentum and energy are conserved. Velocities are u before absorption and V after absorption; then

$$Mu - \frac{hv}{c} e_z = MV \ , \qquad (3)$$

$$\frac{1}{2}Mu^2 + hv = \frac{1}{2}MV^2 + hv_0 \ . \qquad (4)$$

Writing (3) in components, with momentum normal to the z axis conserved, gives

$$Mu\sin\theta = MV\sin\theta' \ , \qquad (5)$$

$$Mu\cos\theta - \frac{hv}{c} = MV\cos\theta' \ , \qquad (6)$$

Here θ' is the excited atom's angle to positive z. From (5), (6) eliminate θ', to obtain

$$M^2u^2 - M^2V^2 = -(\frac{hv}{c})^2 + 2Mu\frac{hv}{c}\cos\theta \ . \qquad (7)$$

From (4), (7) we obtain

$$2hv_0 - 2hv = -\frac{1}{M}(\frac{hv}{c})^2 + 2hv\frac{u}{c}\cos\theta \ . \qquad (8)$$

Using $(hv/c)^2 \square M$ gives

$$v_0 = v(1 + \frac{u}{c}\cos\theta); \qquad (9)$$

In terms of wavelength,

$$\lambda_0 = \frac{\lambda}{1 + \frac{u}{c}\cos\theta} \ . \qquad (10)$$

Solving (10) gives